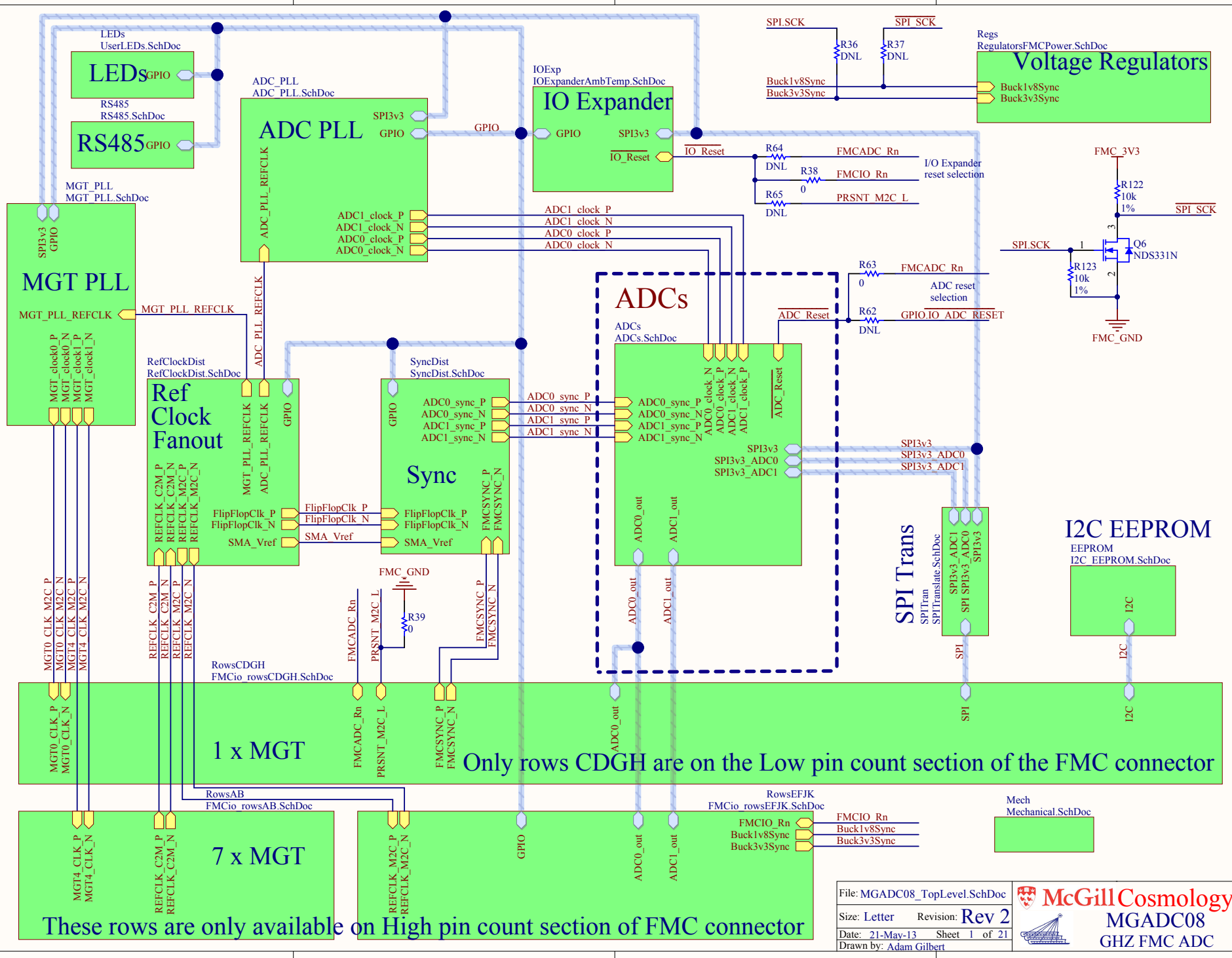
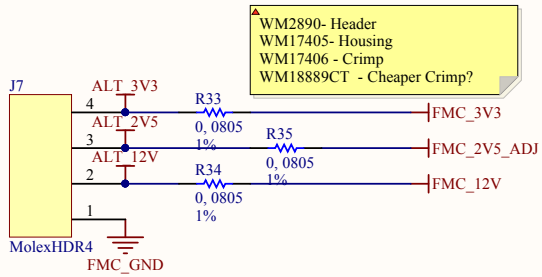
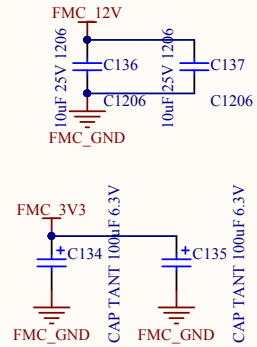
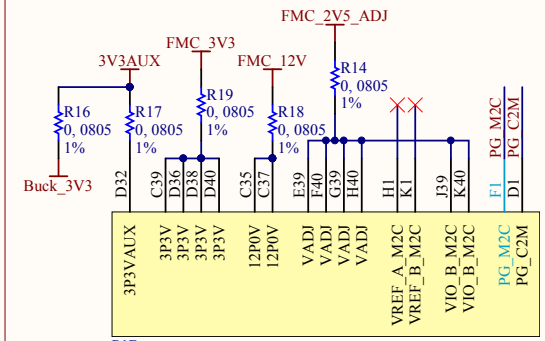

LEDs
UserLEDs.SchDoc
LEDs GPIO
RS485
RS485.SchDoc
RS485 GPIO
MGT_PLL
MGT_PLL.SchDoc
MGT PLL
MGT_PLL_REFCLK
MGT_clock0_P
MGT_clock0_N
MGT_clock1_P
MGT_clock1_N
ADC_PLL
ADC_PLL.SchDoc
ADC PLL
ADC_PLL_REFCLK
SPI3v3
GPIO
ADC1_clock_P
ADC1_clock_N
ADC0_clock_P
ADC0_clock_N
IOExp
IOExpanderAmbTemp.SchDoc
IO Expander
IO_Reset
SPI.SCK
SPI_SCK
R36
DNL
R37
DNL
Buck1v8Sync
Buck3v3Sync
Regs
RegulatorsFMCPower.SchDoc
Voltage Regulators
R64
DNL
R38
0
R65
DNL
FMCADC_Rn
FMCIO_Rn
PRSNT_M2C_L
I/O Expander
reset selection
FMC_3V3
R122
10k
1%
R123
10k
1%
Q6
NDS331N
FMC_GND
RefClockDist
RefClockDist.SchDoc
Ref
Clock
Fanout
REFCLK_C2M_P
REFCLK_C2M_N
REFCLK_M2C_P
REFCLK_M2C_N
FlipFlopClk_P
FlipFlopClk_N
SMA_Vref
SyncDist
SyncDist.SchDoc
Sync
FMCSYNC_P
FMCSYNC_N
ADC0_sync_P
ADC0_sync_N
ADC1_sync_P
ADC1_sync_N
ADCs
ADCs.SchDoc
ADC_Reset
R63
0
R62
DNL
GPIO.IO_ADC_RESET
ADC reset
selection
SPI3v3_ADC0
SPI3v3_ADC1
ADC0_out
ADC1_out
SPI Trans
SPITran
SPITranslate.SchDoc
I2C EEPROM
EEPROM
I2C_EEPROM.SchDoc
I2C
SPI
MGT0_CLK_M2C_P
MGT0_CLK_M2C_N
MGT4_CLK_M2C_P
MGT4_CLK_M2C_N
MGT0_CLK_P
MGT0_CLK_N
RowsCDGH
FMCio_rowsCDGH.SchDoc
1 x MGT
Only rows CDGH are on the Low pin count section of the FMC connector
RowsAB
FMCio_rowsAB.SchDoc
MGT4_CLK_P
MGT4_CLK_N
7 x MGT
RowsEFJK
FMCio_rowsEFJK.SchDoc
These rows are only available on High pin count section of FMC connector
Mech
Mechanical.SchDoc
File: MGADC08_TopLevel.SchDoc
Size: Letter
Revision: Rev 2
Date: 21-May-13
Sheet 1 of 21
Drawn by: Adam Gilbert
McGill Cosmology
MGADC08
GHZ FMC ADC

Alternate Input power (Temporary for power on test)
Pinout compatible with standard fan connector

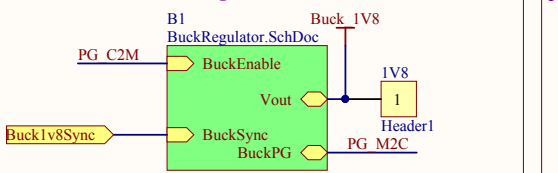




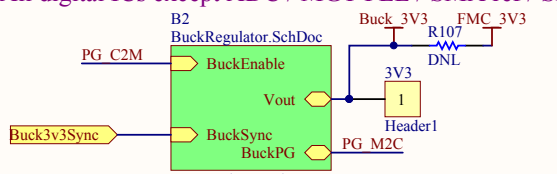
FMC Power Pins

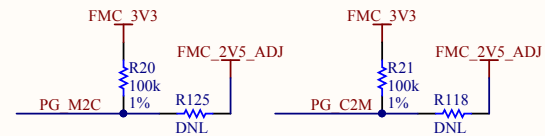


1.8V Regulator from 12V
For ADC SPI and output buffers

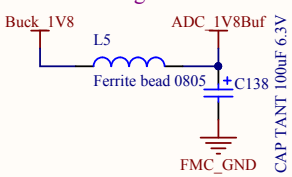


3.3V Regulator from 12V
All digital ICs except ADC / MGT PLL / SMA ref / SMA input

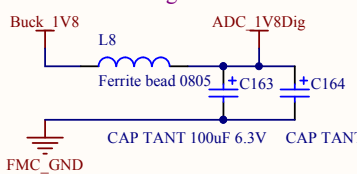




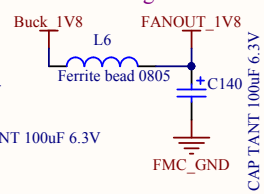
1.8V for ADC output buffers
From buck regulators



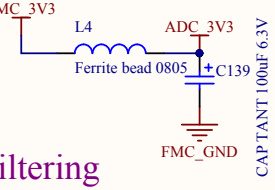
1.8V for ADC SPI core
From buck regulators



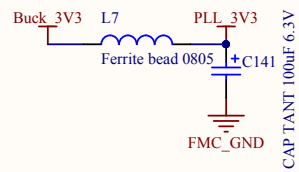
1.8V for Sync and Clock fanout IC
From buck regulators



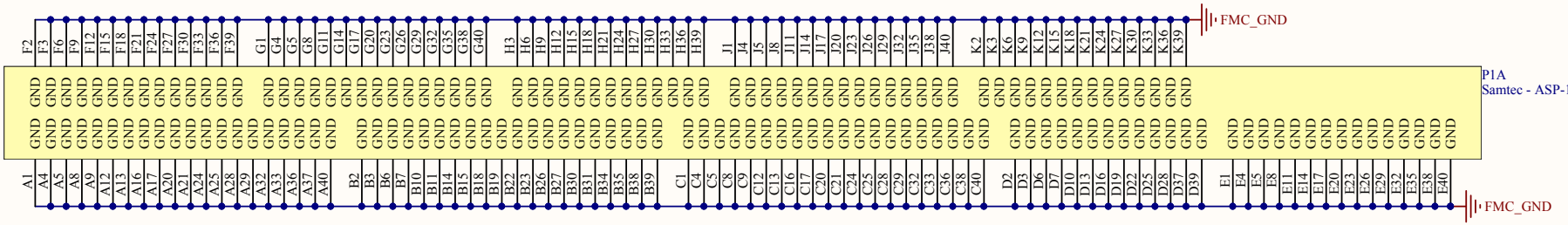
3.3V for ADC Analog core and SPI pads
From FMC



3.3V for ADC PLL
From buck regulators



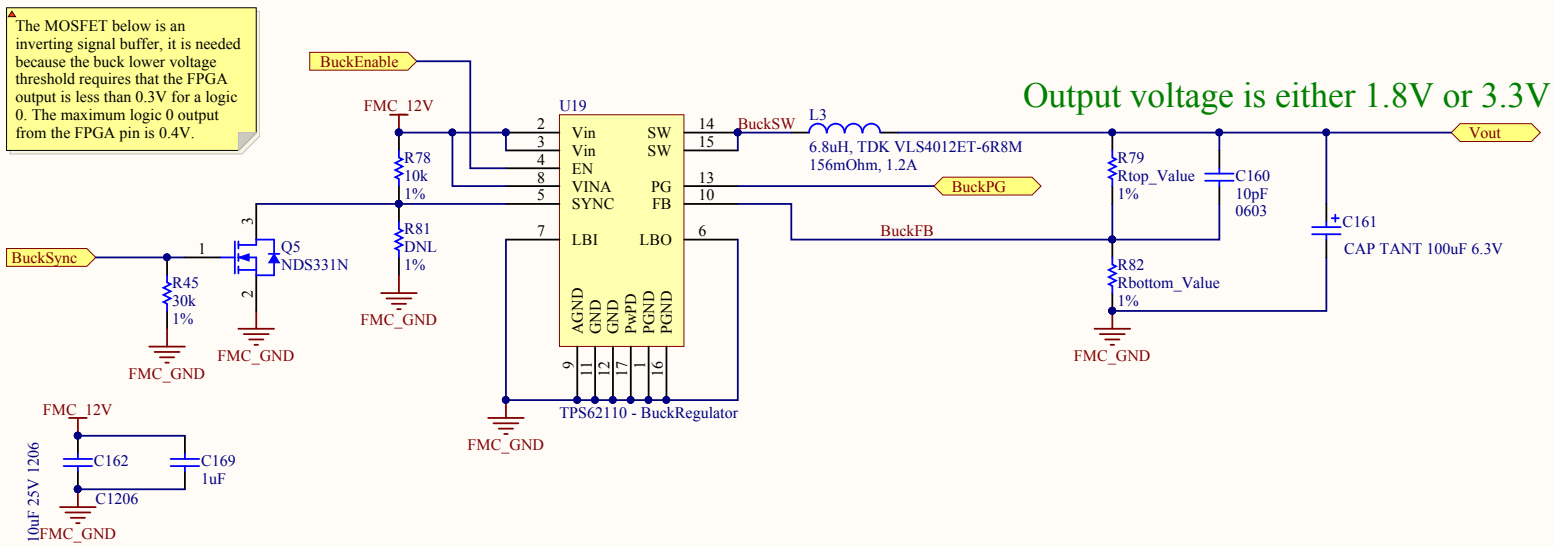
Power Filtering



FMC Ground Pins

File: RegulatorsFMCPower.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 2 of 21
Drawn by: Adam Gilbert	

McGill Cosmology
MGADC08
GHZ FMC ADC



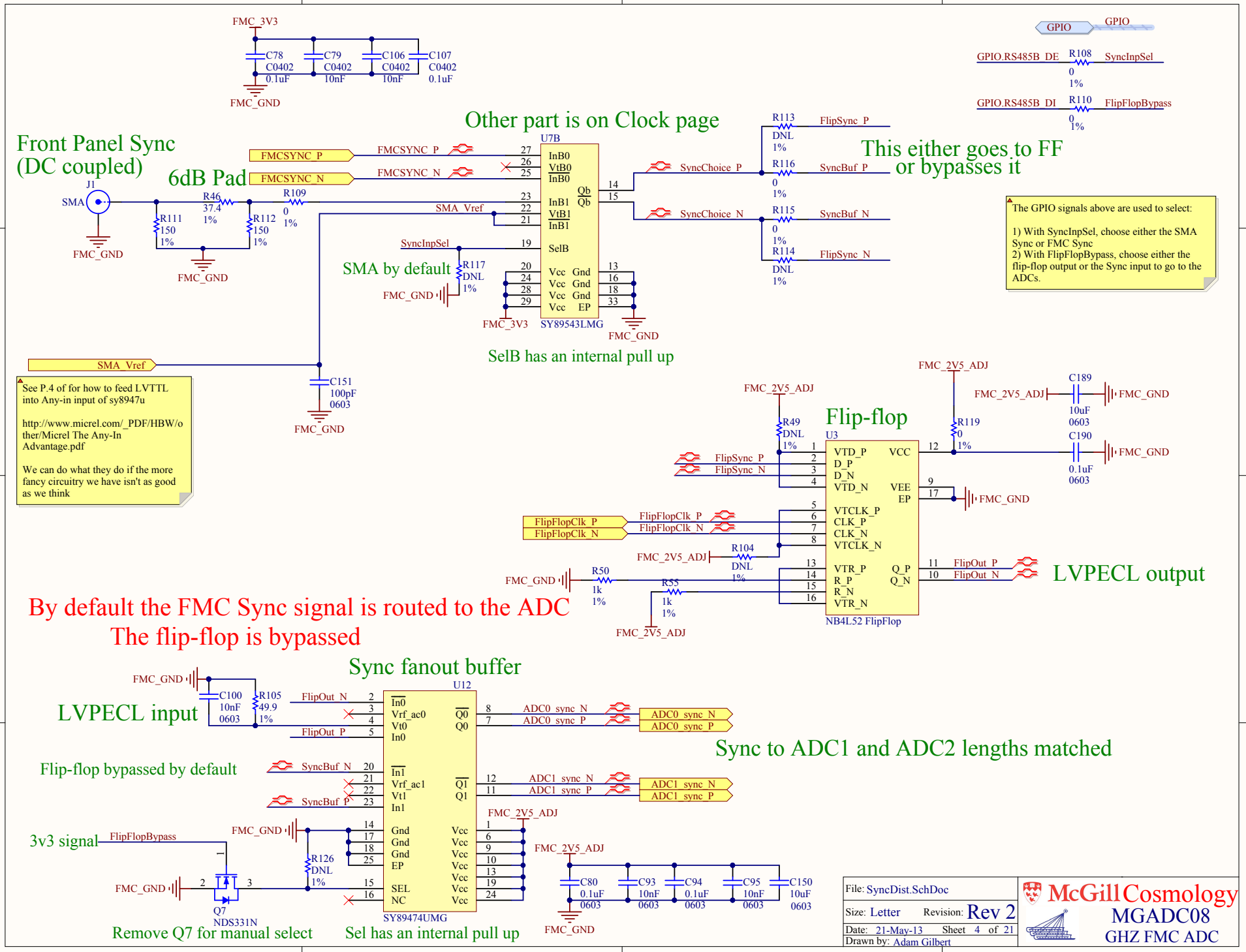
Output voltage is either 1.8V or 3.3V depending on channel

Average output current is approximately 1A, IC Maximum 1.5A
Switching frequency is approximatly 1MHz maximum 1.4Mhz

Output capacitor needs to be >=22uF for stability

File: BuckRegulator.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 3 of 21
Drawn by: Adam Gilbert	

McGill Cosmology
MGADC08
GHZ FMC ADC


FMC_3V3
C78 C0402 0.1uF
C79 C0402 10nF
C106 C0402 10nF
C107 C0402 0.1uF
FMC_GND
GPIO
GPIO.RS485B_DE
R108 0 1%
SyncInpSel
GPIO.RS485B_DI
R110 0 1%
FlipFlopBypass
Other part is on Clock page
Front Panel Sync (DC coupled)
6dB Pad
J1
SMA
R46 37.4 1%
R111 150 1%
R112 150 1%
R109 0 1%
FMCSYNC_P
FMCSYNC_N
SMA_Vref
U7B
InB0
VtB0
InB0
InB1
VtB1
InB1
SelB
Qb
Qb
Vcc Gnd
Vcc Gnd
Vcc Gnd
Vcc EP
SyncChoice_P
SyncChoice_N
R113 DNL 1%
R116 0 1%
R115 0 1%
R114 DNL 1%
FlipSync_P
SyncBuf_P
SyncBuf_N
FlipSync_N
This either goes to FF or bypasses it
SyncInpSel
SMA by default
R117 DNL 1%
SY89543LMG
SelB has an internal pull up
The GPIO signals above are used to select:
1) With SyncInpSel, choose either the SMA Sync or FMC Sync
2) With FlipFlopBypass, choose either the flip-flop output or the Sync input to go to the ADCs.
SMA_Vref
C151 100pF 0603
See P.4 of for how to feed LVTTL into Any-in input of sy8947u
http://www.micrel.com/_PDF/HBW/other/Micrel The Any-In Advantage.pdf
We can do what they do if the more fancy circuitry we have isn't as good as we think
Flip-flop
U3
VTD_P
D_P
D_N
VTD_N
VCC
VEE
EP
VTCLK_P
CLK_P
CLK_N
VTCLK_N
VTR_P
R_P
R_N
VTR_N
Q_P
Q_N
R49 DNL 1%
R119 0 1%
C189 10uF 0603
C190 0.1uF 0603
R104 DNL 1%
R50 1k 1%
R55 1k 1%
FlipOut_P
FlipOut_N
NB4L52 FlipFlop
LVPECL output
By default the FMC Sync signal is routed to the ADC
The flip-flop is bypassed
Sync fanout buffer
U12
In0
Vrf_ac0
Vt0
In0
In1
Vrf_ac1
Vt1
In1
Gnd
Vcc
EP
SEL
NC
Q0
Q1
ADC0_sync_N
ADC0_sync_P
ADC1_sync_N
ADC1_sync_P
LVPECL input
C100 10nF 0603
R105 49.9 1%
Flip-flop bypassed by default
SyncBuf_N
SyncBuf_P
Sync to ADC1 and ADC2 lengths matched
3v3 signal
FlipFlopBypass
R126 DNL 1%
Q7 NDS331N
SY89474UMG
Remove Q7 for manual select
Sel has an internal pull up
C80 0.1uF 0603
C93 10nF 0603
C94 0.1uF 0603
C95 10nF 0603
C150 10uF 0603
File: SyncDist.SchDoc
Size: Letter
Revision: Rev 2
Date: 21-May-13
Sheet 4 of 21
Drawn by: Adam Gilbert
McGill Cosmology
MGADC08
GHZ FMC ADC

Can emulate front panel clock with this

Other part is on Sync page

-6dB Pad

Clock input on Front Panel
(AC coupled)

SMA by default

SelA has an internal pull up

1.4V

Opamp used as voltage sink/source

ADC_PLL_REFCLK trace Zc=100 Ohms

The GPIO signal above are used to select either the SMA Clock or FMC Clock

See P.4 of for how to feed LVTTL into Any-in input

http://www.micrel.com/_PDF/HBW/other/Micrel The Any-In Advantage.pdf

See Figure 27 of http://www.analog.com/static/imported-files/data_sheets/ADCLK846.pdf For CMOS output termination

The ground plane nearest the ADC_PLL_REFCLK, when the trace is both on the top or bottom copper has been cut away. This gives a 17.38mil spacing between the trace and its ground plane. A top copper trace width of 6.5mils over the ground plane cut out region gives 100 Ohms

Iavg going into Pin.21/22 above is 0mA

CONN, Oscillator Socket, Half Size

sy89543L

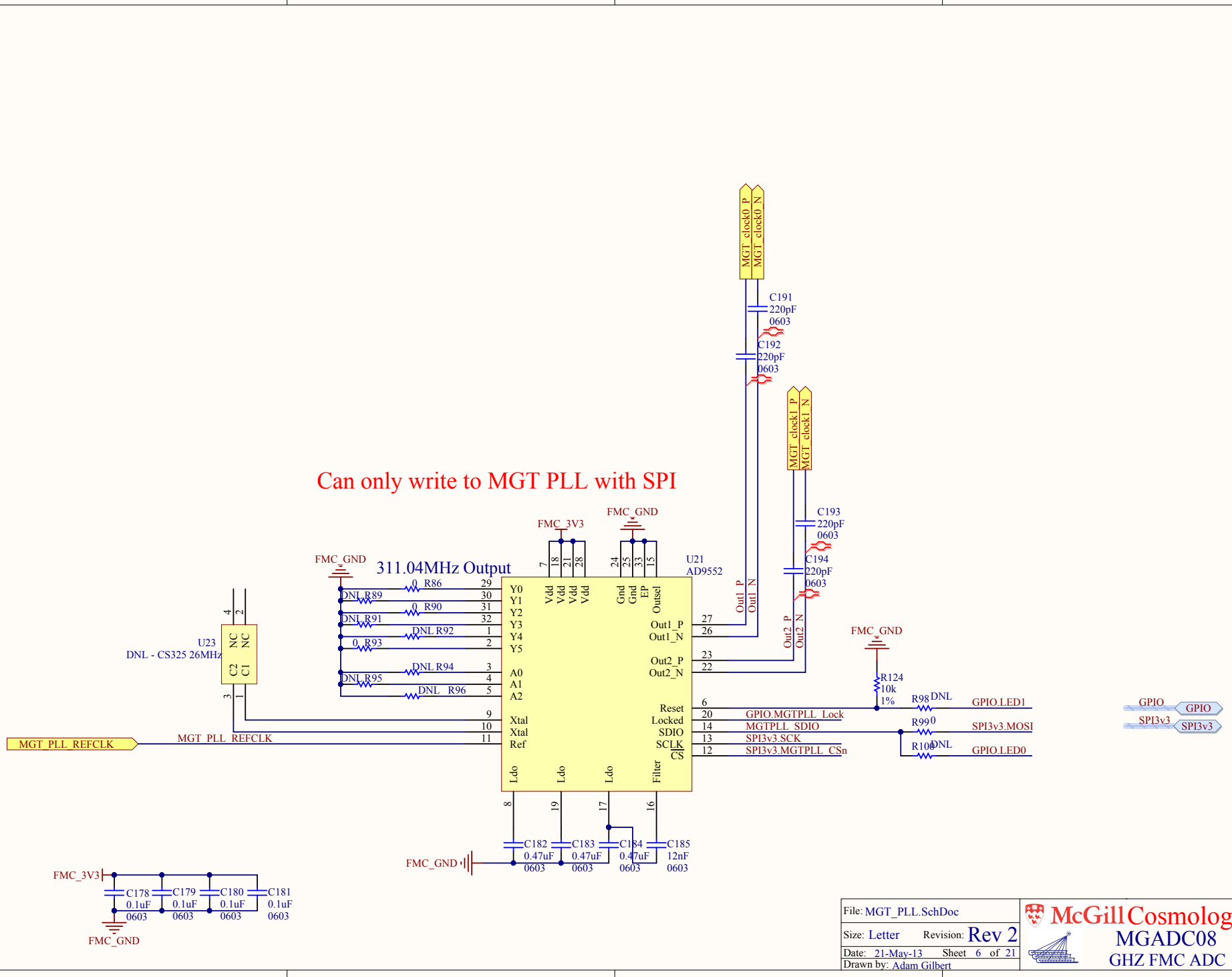
ADCLK846

Lm8261

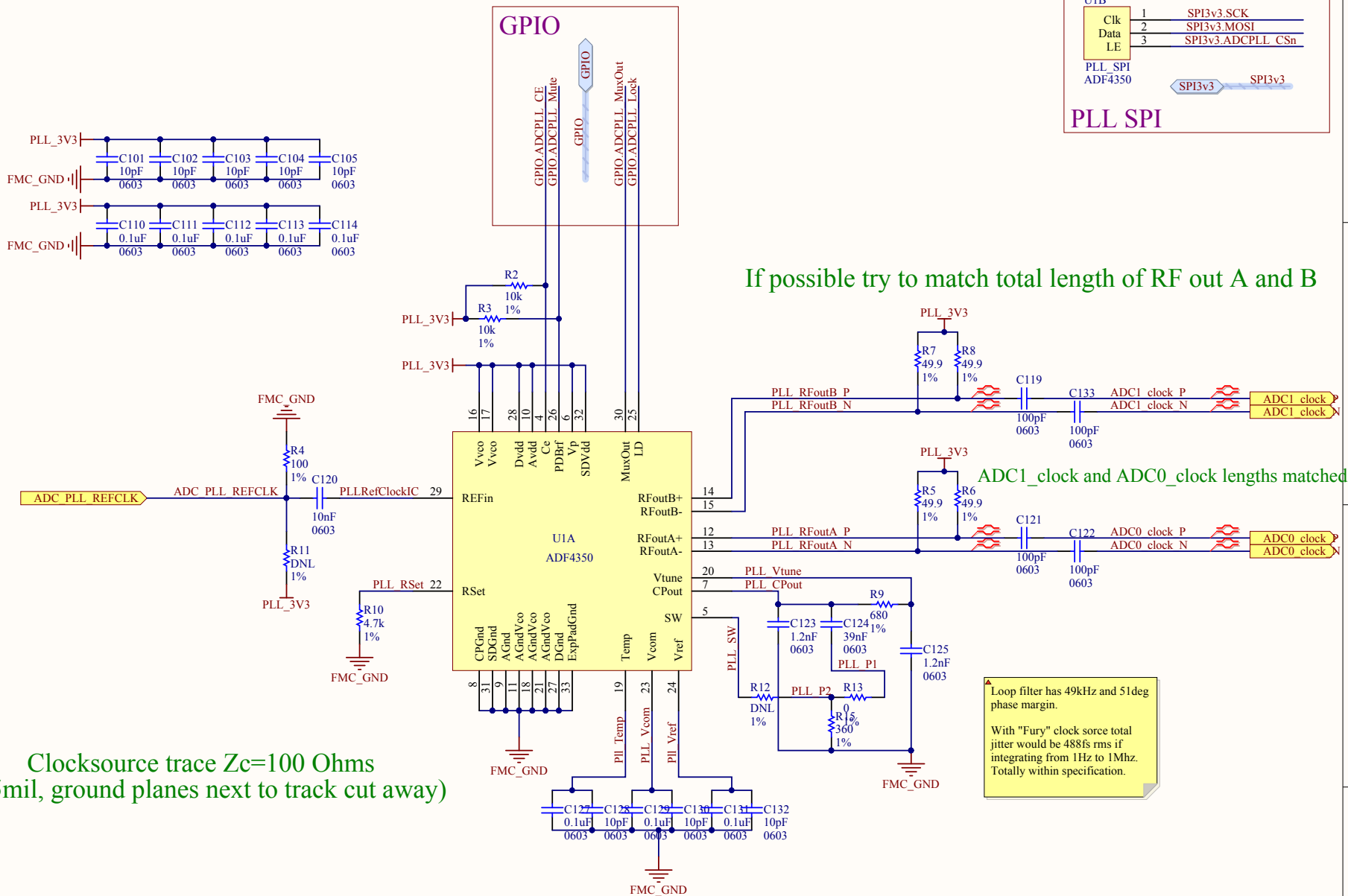
Configured as LVDS

Configured as CMOS

File: RefClockDist.SchDoc	McGill Cosmology
Size: Letter Revision: Rev 2	MGADC08
Date: 21-May-13 Sheet 5 of 21	GHZ FMC ADC
Drawn by: Adam Gilbert	

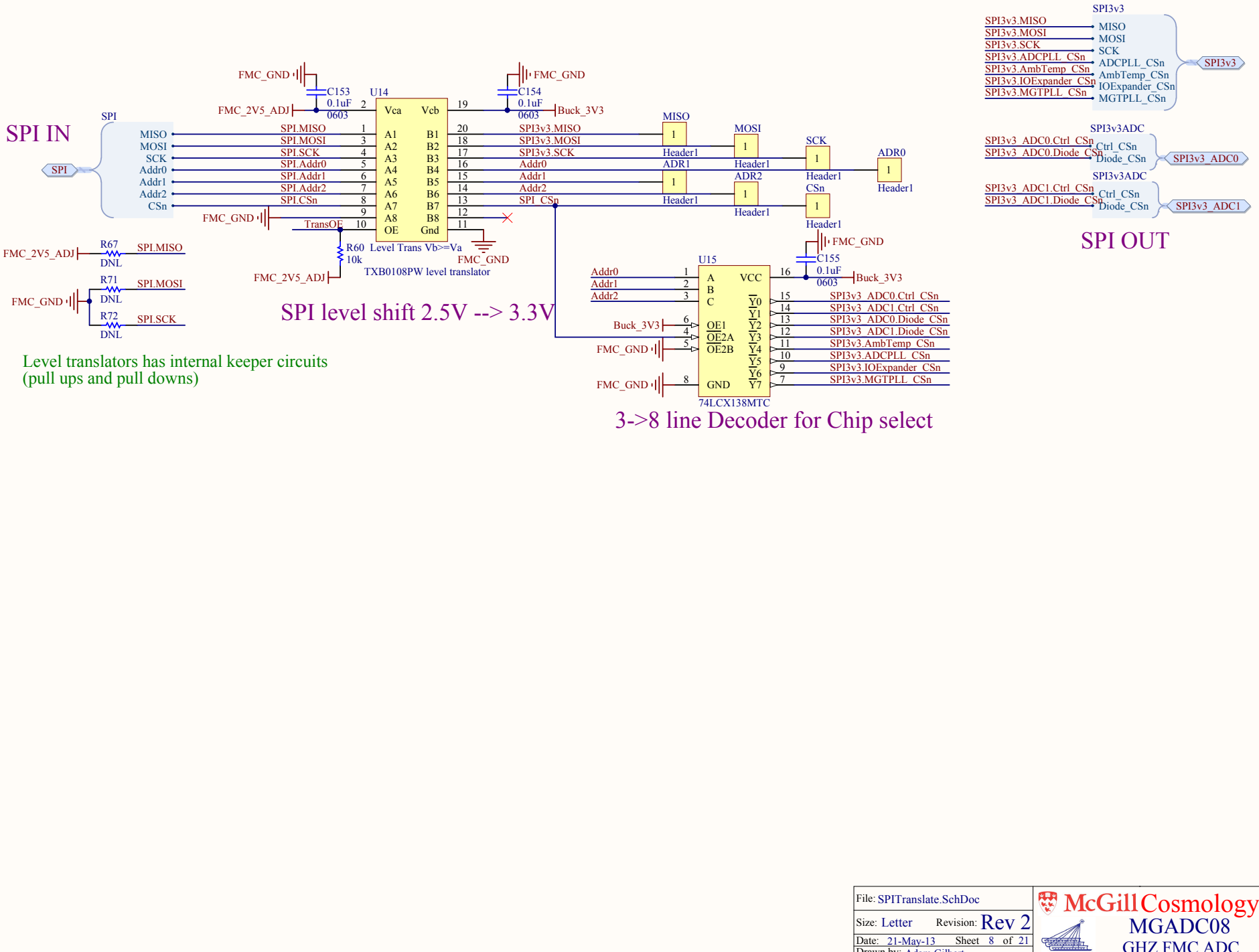

Can only write to MGT PLL with SPI
311.04MHz Output
U21
AD9552
U23
DNL - CS325 26MHz
MGT_clock0_P
MGT_clock0_N
MGT_clock1_P
MGT_clock1_N
C191 220pF 0603
C192 220pF 0603
C193 220pF 0603
C194 220pF 0603
FMC_3V3
FMC_GND
MGT_PLL_REFCLK
R124 10k 1%
R98 DNL
R99 0
R100 DNL
GPIO.LED1
SPI3v3.MOSI
GPIO.LED0
GPIO.MGTPLL_Lock
MGTPLL_SDIO
SPI3v3.SCK
SPI3v3.MGTPLL_CSn
C182 0.47uF 0603
C183 0.47uF 0603
C184 0.47uF 0603
C185 12nF 0603
C178 0.1uF 0603
C179 0.1uF 0603
C180 0.1uF 0603
C181 0.1uF 0603
File: MGT_PLL.SchDoc
Size: Letter Revision: Rev 2
Date: 21-May-13 Sheet 6 of 21
Drawn by: Adam Gilbert
McGill Cosmology
MGADC08
GHZ FMC ADC

Please place crystal on the bottom of the board - socket will be removed after prototyping



File: ADC_PLL.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 7 of 21
Drawn by: Adam Gilbert	

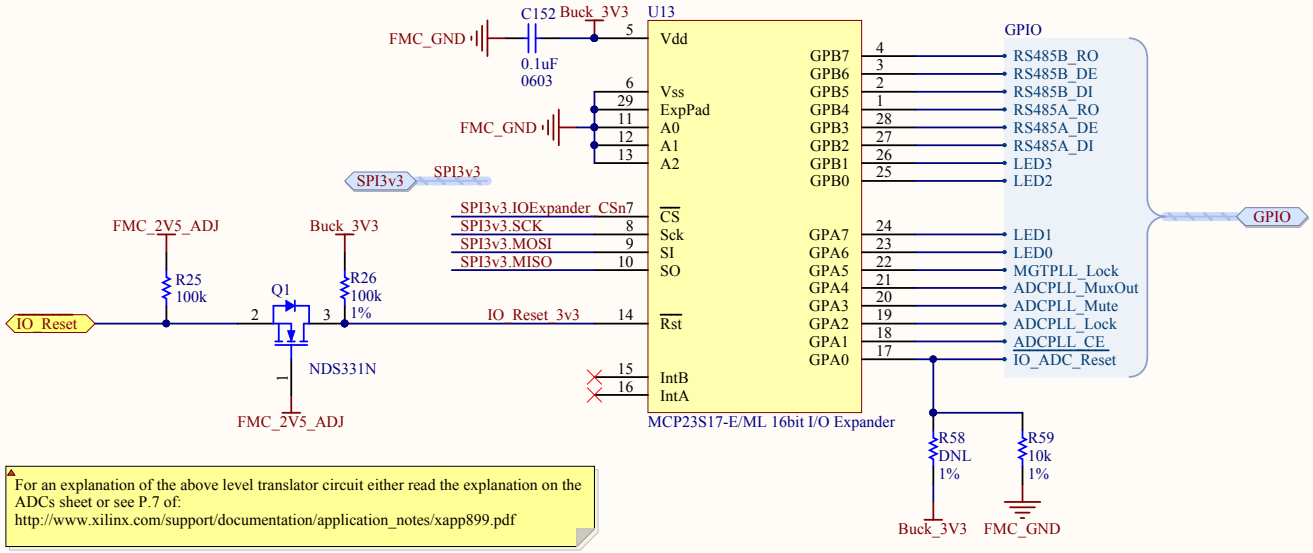
McGill Cosmology
MGADC08
GHZ FMC ADC



File: SPITranslate.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 8 of 21
Drawn by: Adam Gilbert	

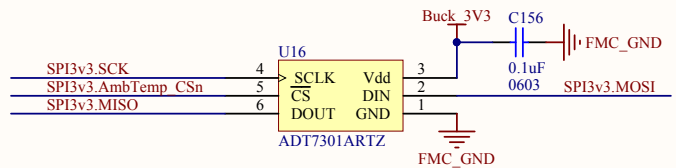
McGill Cosmology
MGADC08
GHZ FMC ADC

I/O expander

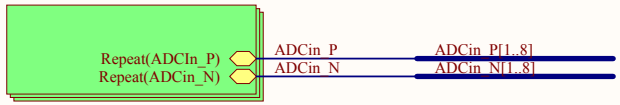


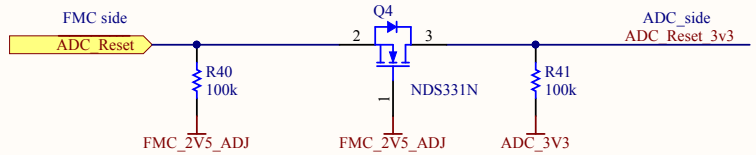
For an explanation of the above level translator circuit either read the explanation on the ADCs sheet or see P.7 of:
http://www.xilinx.com/support/documentation/application_notes/xapp899.pdf

Ambient temperature



File: IOExpanderAmbTemp.SchDoc	McGill Cosmology
Size: Letter Revision: Rev 2	MGADC08
Date: 21-May-13 Sheet 9 of 21	GHZ FMC ADC
Drawn by: Adam Gilbert	

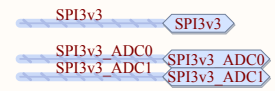




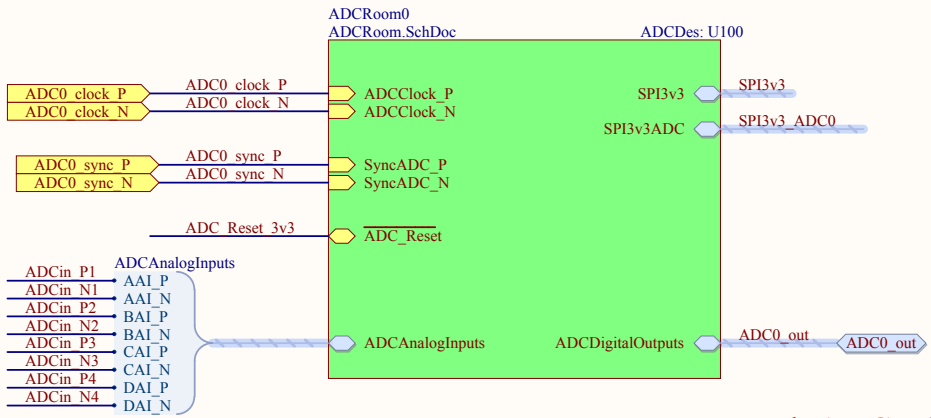
If the ADC_reset on the FMC side is grounded(0V) the MOSFET will turn on and ADC_reset on the ADC side will be pulled to 0V.

If the ADC_reset on the FMC side is logic high (2.5V), the MOSFET will turn off and the ADC_reset on the ADC side will float up to 3.3V via the pull-up.

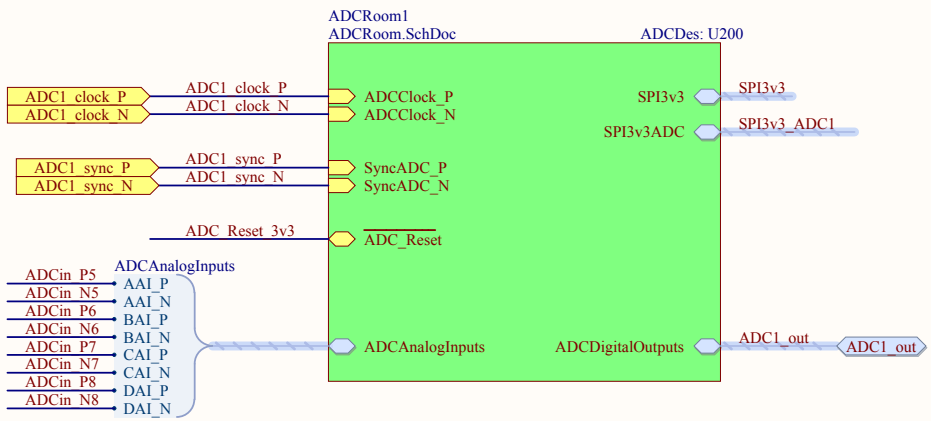
This is actually a bi-directional level translator although we're not using it like that (think about the diode when looking at it in the other direction)



Sync lines to each ADC are the same length

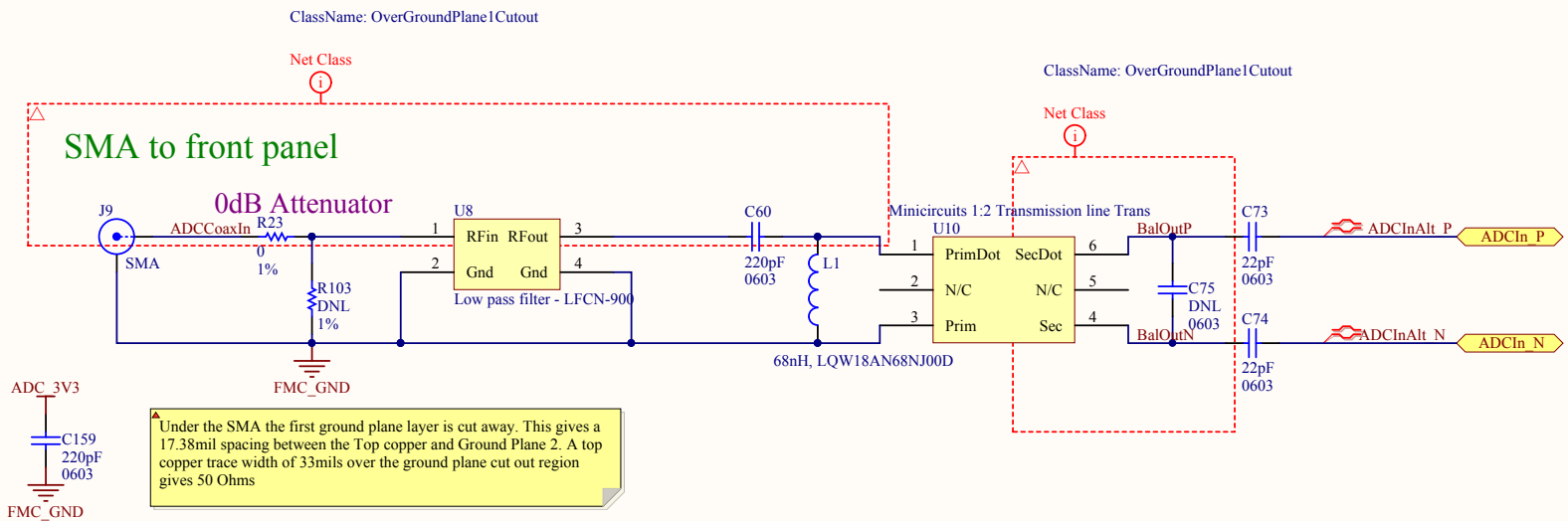


Both ADCs share Reset



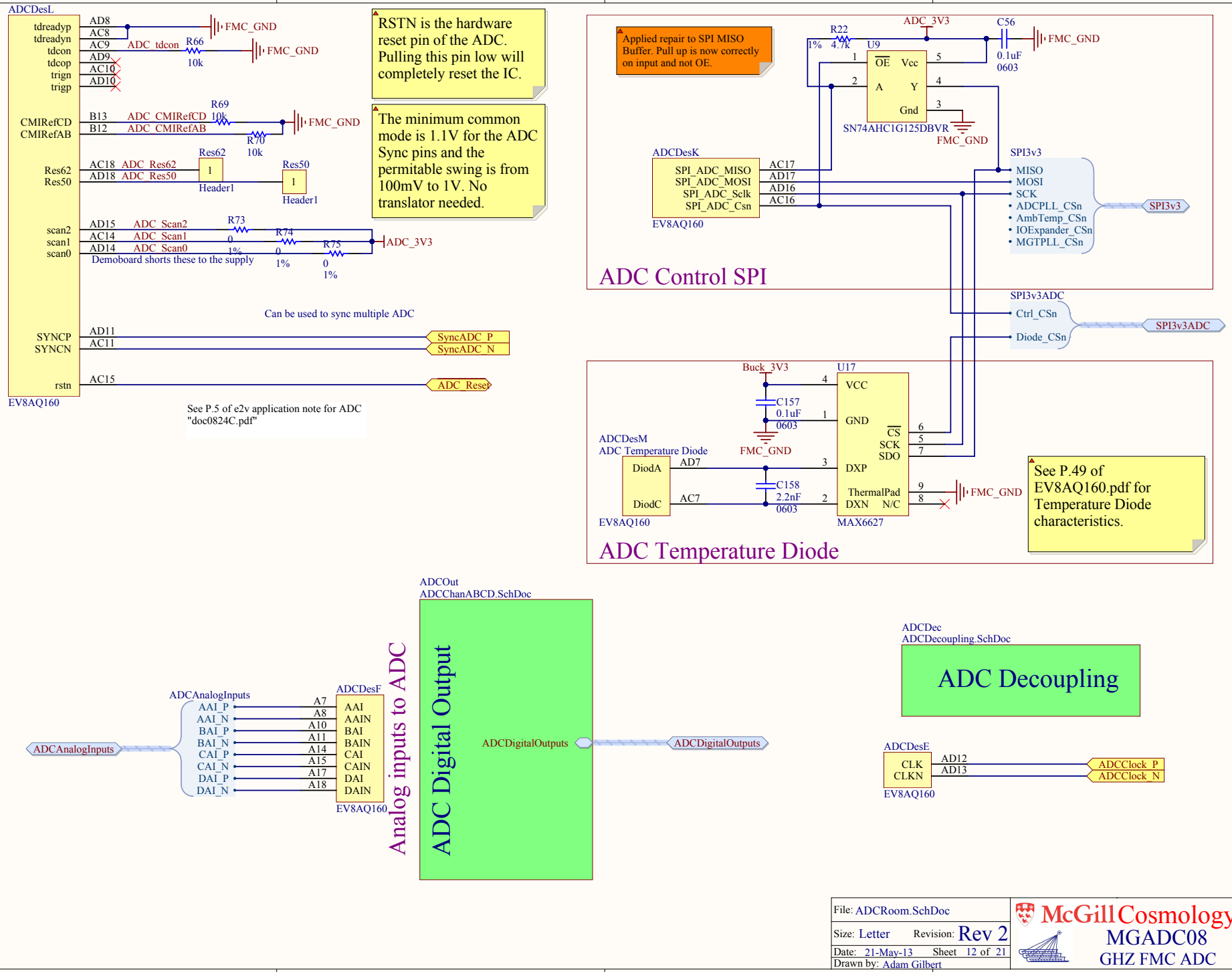
File: ADCs.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 10 of 21
Drawn by: Adam Gilbert	

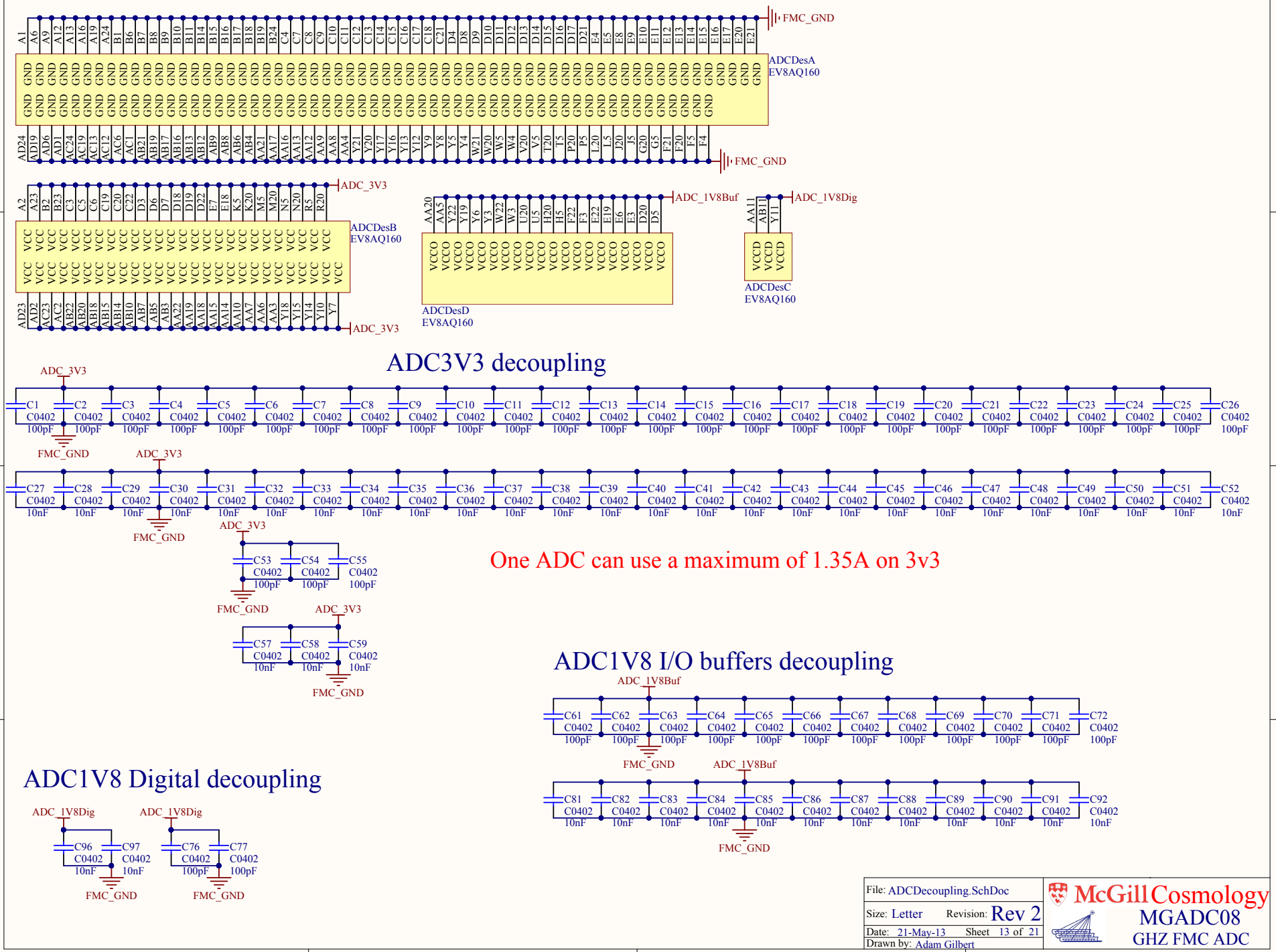
McGill Cosmology
MGADC08
GHZ FMC ADC



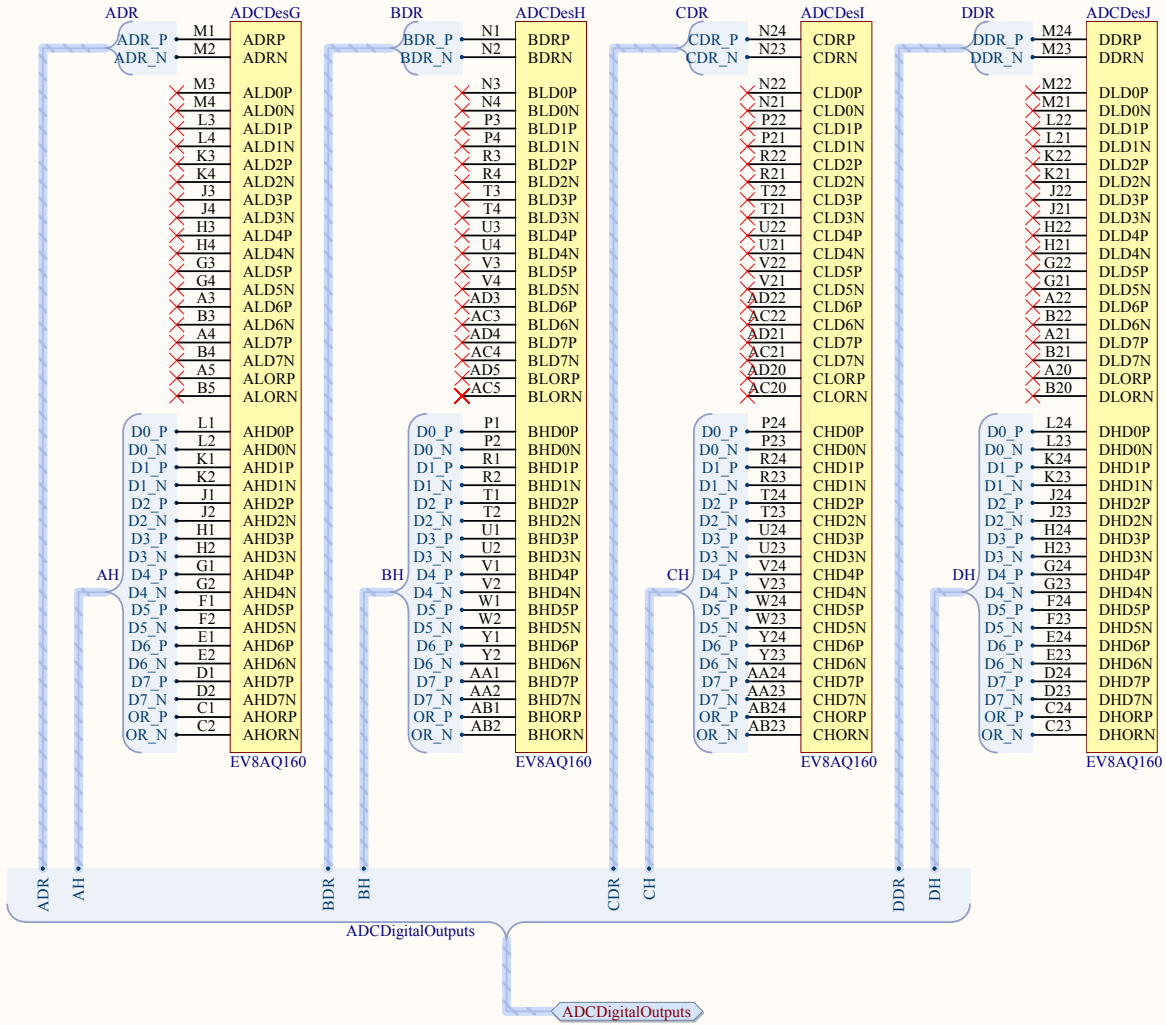
File: AnalogInputChannel.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 11 of 21
Drawn by: Adam Gilbert	

McGill Cosmology
MGADC08
GHZ FMC ADC


ADCDesL
tdreadyp
tdreadyn
tdcon
tdcop
trign
trigp
AD8
AC8
AC9
AD9
AC10
AD10
ADC_tdcon
R66
10k
FMC_GND
CMIRefCD
CMIRefAB
B13
B12
ADC_CMIRefCD
ADC_CMIRefAB
R69
10k
R70
10k
Res62
Res50
AC18
AD18
ADC_Res62
ADC_Res50
Header1
scan2
scan1
scan0
AD15
AC14
AD14
ADC_Scan2
ADC_Scan1
ADC_Scan0
R73
R74
R75
0
1%
ADC_3V3
Demoboard shorts these to the supply
Can be used to sync multiple ADC
SYNCP
SYNCN
AD11
AC11
SyncADC_P
SyncADC_N
rstn
AC15
ADC_Reset
EV8AQ160
See P.5 of e2v application note for ADC "doc0824C.pdf"
RSTN is the hardware reset pin of the ADC. Pulling this pin low will completely reset the IC.
The minimum common mode is 1.1V for the ADC Sync pins and the permitable swing is from 100mV to 1V. No translator needed.
Applied repair to SPI MISO Buffer. Pull up is now correctly on input and not OE.
R22
4.7k
U9
OE
Vcc
A
Y
Gnd
SN74AHC1G125DBVR
C56
0.1uF
0603
ADC_3V3
FMC_GND
ADCDesK
SPI_ADC_MISO
SPI_ADC_MOSI
SPI_ADC_Sclk
SPI_ADC_Csn
AC17
AD17
AD16
AC16
EV8AQ160
SPI3v3
MISO
MOSI
SCK
ADCPLL_CSn
AmbTemp_CSn
IOExpander_CSn
MGTPLL_CSn
ADC Control SPI
SPI3v3ADC
Ctrl_CSn
Diode_CSn
Buck_3V3
U17
VCC
GND
CS
SCK
SDO
DXP
ThermalPad
DXN
N/C
C157
0.1uF
0603
C158
2.2nF
0603
ADCDesM
ADC Temperature Diode
DiodA
DiodC
AD7
AC7
MAX6627
See P.49 of EV8AQ160.pdf for Temperature Diode characteristics.
ADC Temperature Diode
ADCOut
ADCChanABCD.SchDoc
ADC Digital Output
ADCDigitalOutputs
Analog inputs to ADC
ADCAnalogInputs
AAI_P
AAI_N
BAI_P
BAI_N
CAI_P
CAI_N
DAI_P
DAI_N
ADCDesF
AAI
AAIN
BAI
BAIN
CAI
CAIN
DAI
DAIN
EV8AQ160
ADCDec
ADCDecoupling.SchDoc
ADC Decoupling
ADCDesE
CLK
CLKN
AD12
AD13
ADCClock_P
ADCClock_N
File: ADCRoom.SchDoc
Size: Letter
Revision: Rev 2
Date: 21-May-13
Sheet 12 of 21
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FMC_GND
ADCDesA
EV8AQ160
ADC_3V3
ADCDesB
EV8AQ160
ADC_1V8Buf
ADC_1V8Dig
ADCDesD
EV8AQ160
ADCDesC
EV8AQ160
ADC3V3 decoupling
One ADC can use a maximum of 1.35A on 3v3
ADC1V8 I/O buffers decoupling
ADC1V8 Digital decoupling
File: ADCDecoupling.SchDoc
Size: Letter
Revision: Rev 2
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Digital pins of ADC all need to connect to FMC



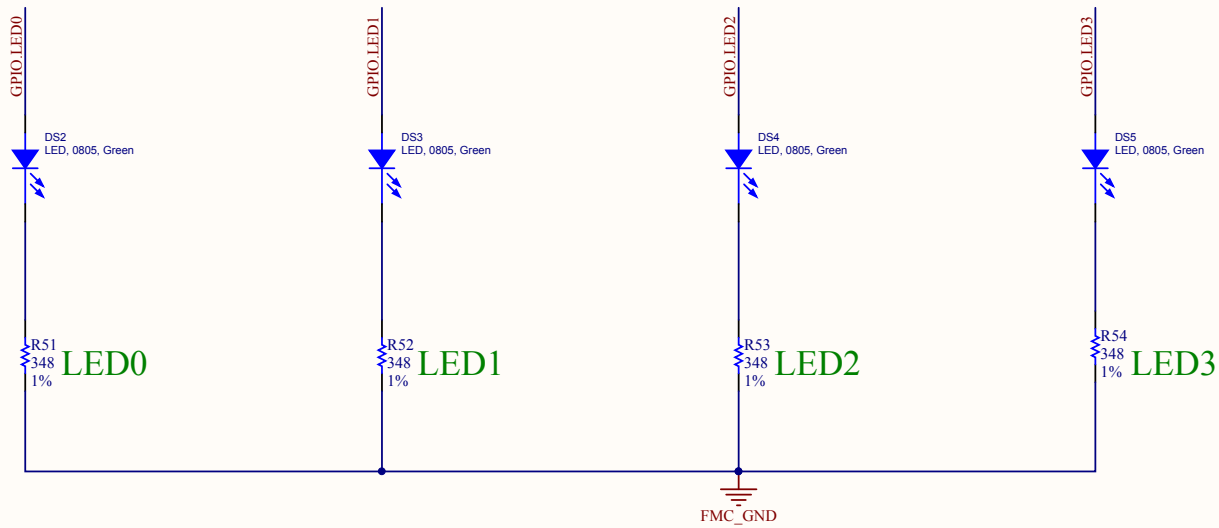
In 1:1 DMux mode, L pins can float, H pins are output,
see P.5 of "EV8AO160.pdf"
See P.3 of "EV8AQ160.pdf"

File: ADCChanABCD.SchDoc	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet 14 of 21
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User LEDs

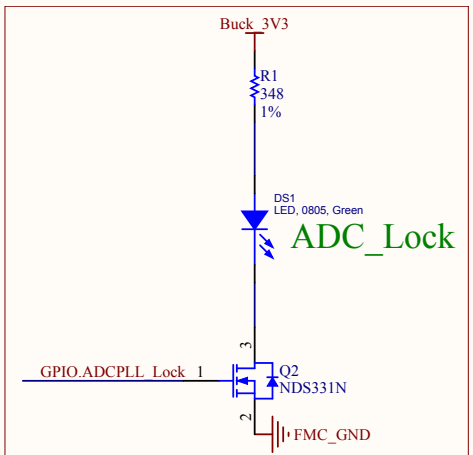
The IO expander can drive 25mA into each output. LED requires 4mA



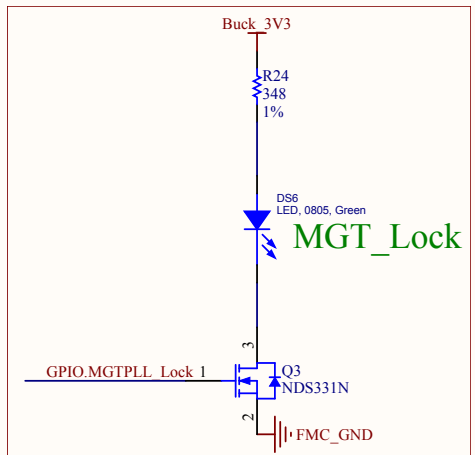
GPIO GPIO

Power indicator





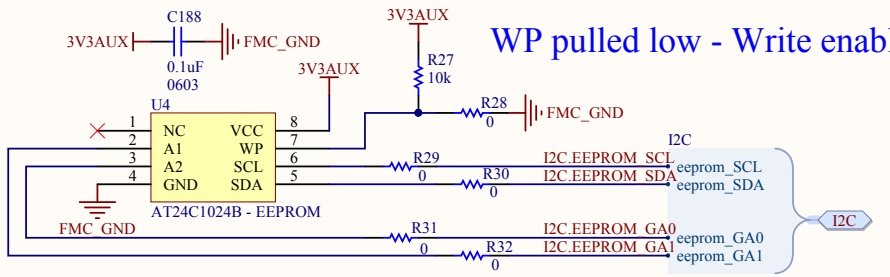
ADC PLL Lock detect LED



MGT PLL Lock detect LED

File: UserLEDs.SchDoc	McGill Cosmology
Size: Letter Revision: Rev 2	MGADC08
Date: 21-May-13 Sheet 15 of 21	GHZ FMC ADC
Drawn by: Adam Gilbert	

http://www.atmel.com/dyn/resources/prod_documents/doc5194.pdf



WP pulled low - Write enabled

Vita FMC spec Pages 63 to 67 discuss the format of what must be stored in the EEPROM and how the EEPROM should be wired

Rule 5.70 specifies that EEPROM is powered by 3.3VAUX

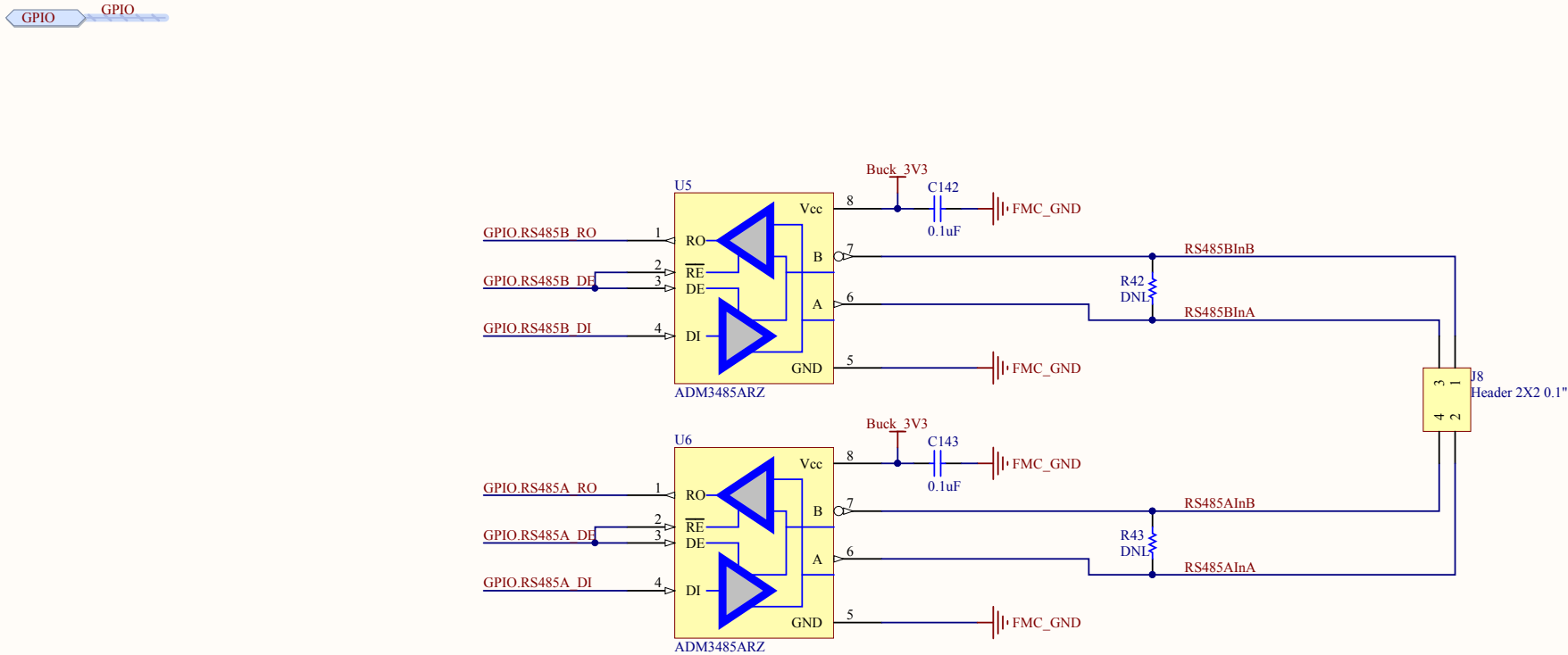
ML605 - HPC FMC

GA0 = LOW and GA1=LOW
I2C Address = 0x50 for Page 0 (64kB per page)
0x51 for Page 1
Level translator present from 3V3 logic to 2V5 logic on ML605

SP605 - HPC FMC

GA0 =HIGH and GA1=LOW
No level translator present on SP605

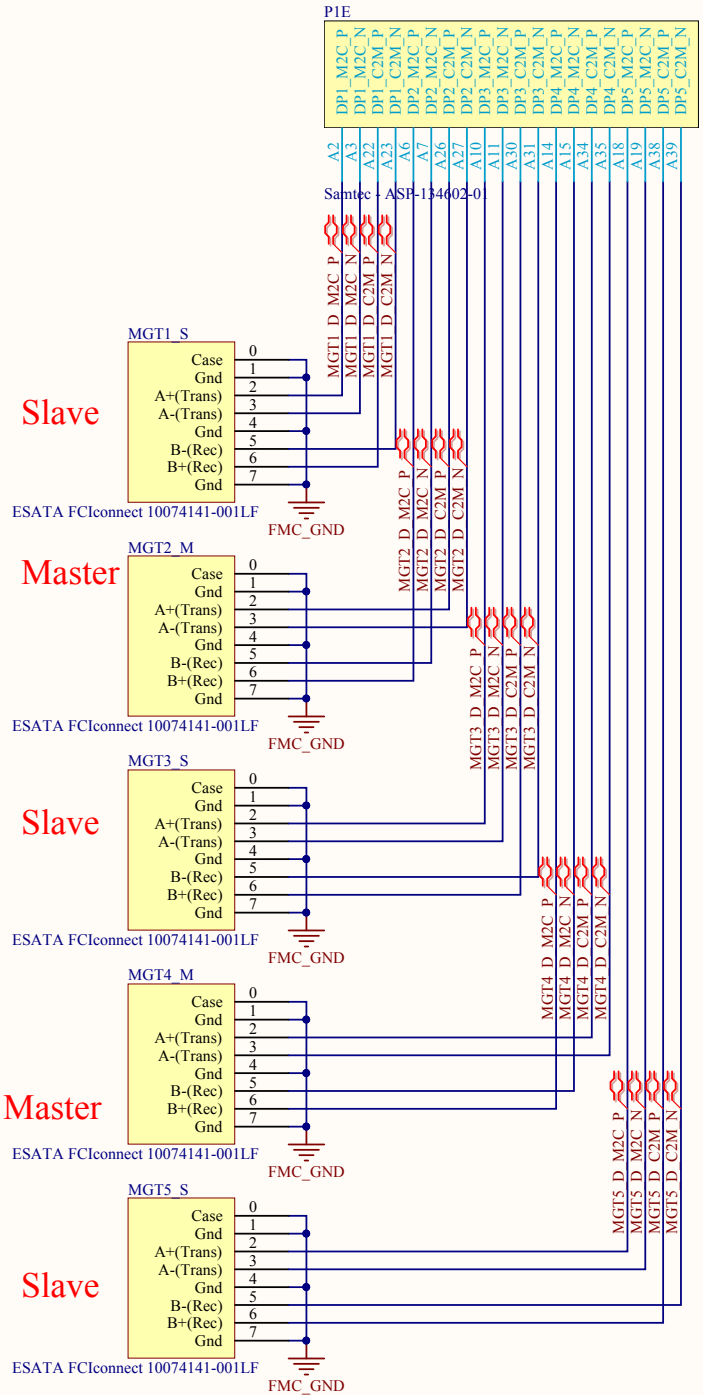
File: I2C_EEPROM.SchDoc		McGill Cosmology
Size: Letter	Revision: Rev 2	MGADC08
Date: 21-May-13	Sheet 16 of 21	GHZ FMC ADC
Drawn by: Adam Gilbert		

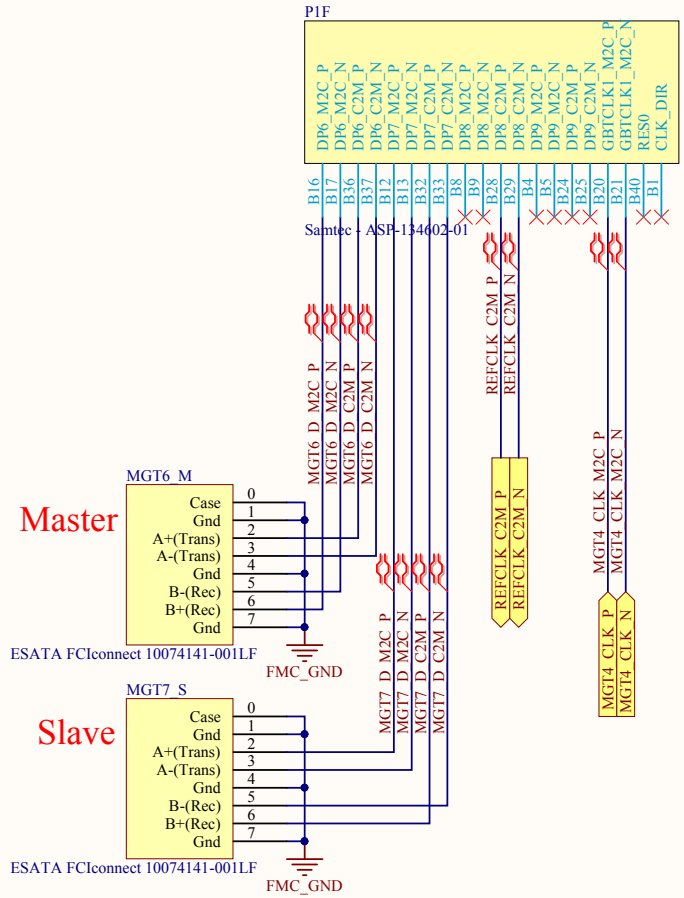


File: RS485.SchDoc	
Size: Letter	Revision: Rev 2
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MGADC08
GHZ FMC ADC

Half of eSATAs are Slave and half are Master

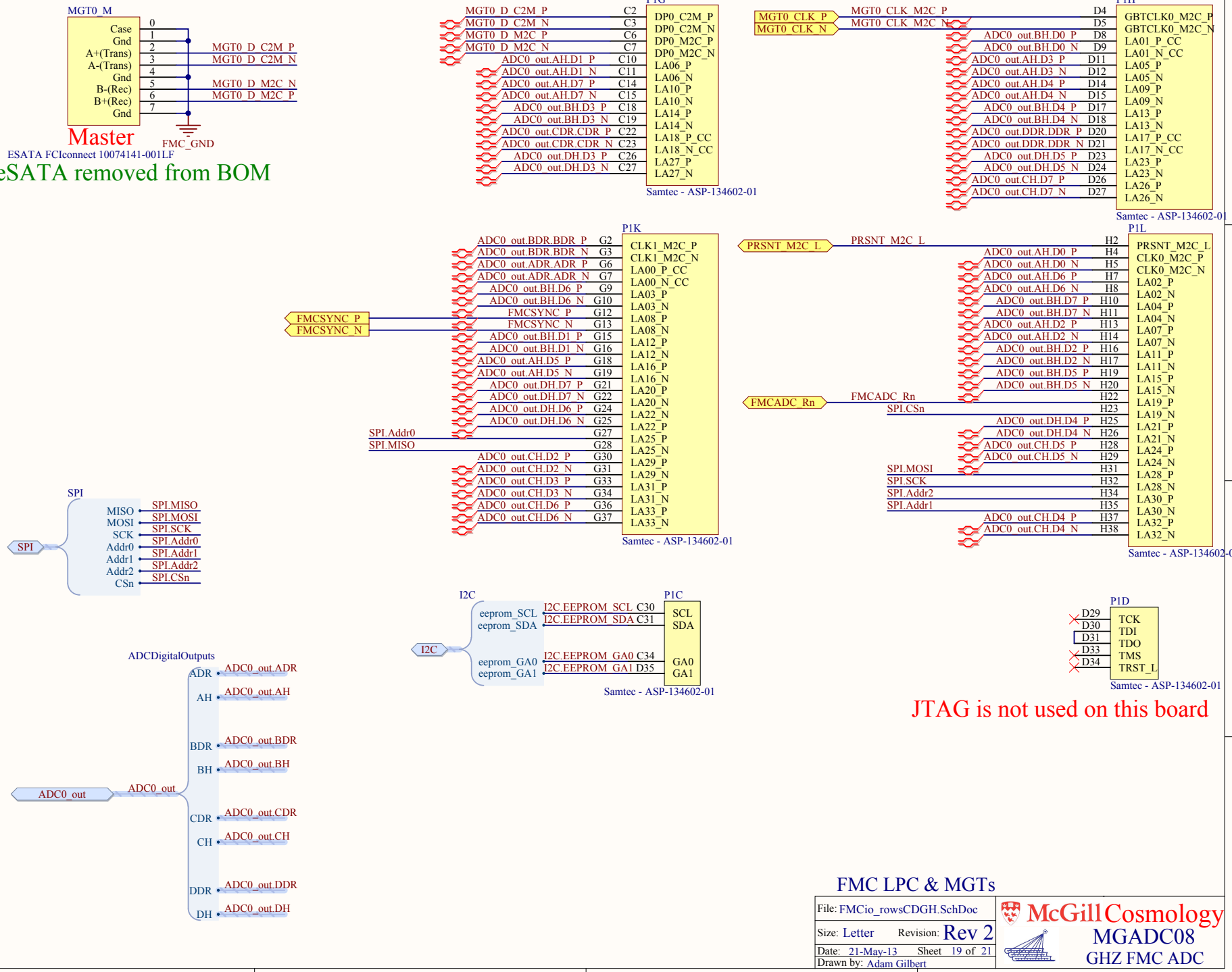


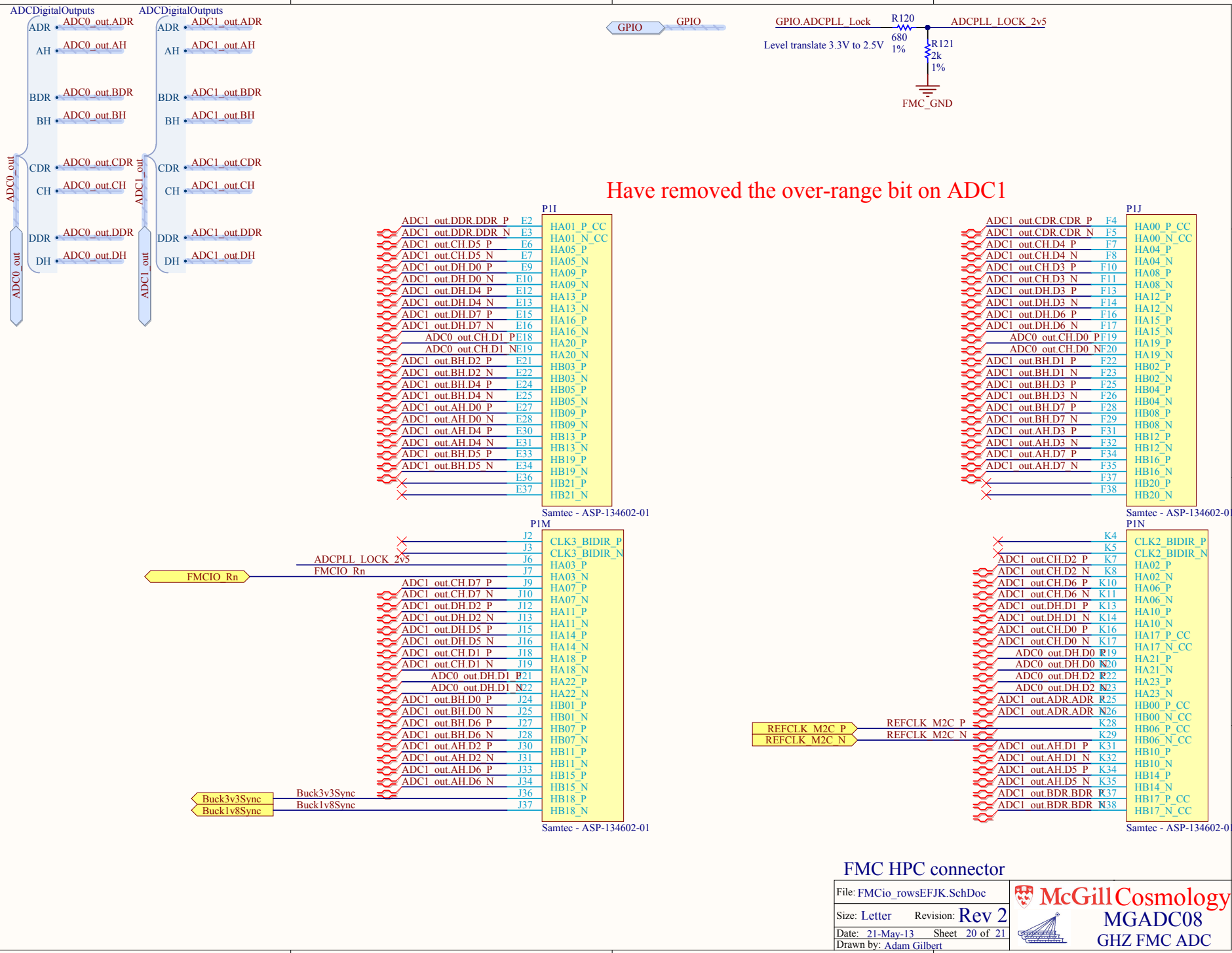


These eSATA connectors will be non FMC compliant
eSATA removed from BOM

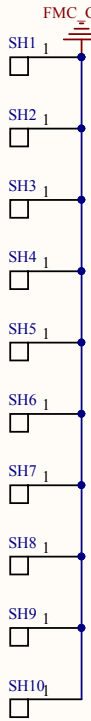
FMC HPC & MGT

File: FMCio_rowsAB.SchDoc	McGill Cosmology
Size: Letter Revision: Rev 2	MGADC08
Date: 21-May-13 Sheet 18 of 21	GHZ FMC ADC
Drawn by: Adam Gilbert	

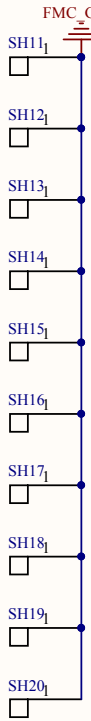

MGT0_M
Case 0
Gnd 1
A+(Trans) 2
A-(Trans) 3
Gnd 4
B-(Rec) 5
B+(Rec) 6
Gnd 7
MGT0_D_C2M_P
MGT0_D_C2M_N
MGT0_D_M2C_N
MGT0_D_M2C_P
FMC_GND
Master
ESATA FCIconnect 10074141-001LF
eSATA removed from BOM
P1G
MGT0_D_C2M_P C2
MGT0_D_C2M_N C3
MGT0_D_M2C_P C6
MGT0_D_M2C_N C7
ADC0_out.AH.D1_P C10
ADC0_out.AH.D1_N C11
ADC0_out.AH.D7_P C14
ADC0_out.AH.D7_N C15
ADC0_out.BH.D3_P C18
ADC0_out.BH.D3_N C19
ADC0_out.CDR.CDR_P C22
ADC0_out.CDR.CDR_N C23
ADC0_out.DH.D3_P C26
ADC0_out.DH.D3_N C27
DP0_C2M_P
DP0_C2M_N
DP0_M2C_P
DP0_M2C_N
LA06_P
LA06_N
LA10_P
LA10_N
LA14_P
LA14_N
LA18_P_CC
LA18_N_CC
LA27_P
LA27_N
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P1H
MGT0_CLK_P
MGT0_CLK_N
MGT0_CLK_M2C_P D4
MGT0_CLK_M2C_N D5
ADC0_out.BH.D0_P D8
ADC0_out.BH.D0_N D9
ADC0_out.AH.D3_P D11
ADC0_out.AH.D3_N D12
ADC0_out.AH.D4_P D14
ADC0_out.AH.D4_N D15
ADC0_out.BH.D4_P D17
ADC0_out.BH.D4_N D18
ADC0_out.DDR.DDR_P D20
ADC0_out.DDR.DDR_N D21
ADC0_out.DH.D5_P D23
ADC0_out.DH.D5_N D24
ADC0_out.CH.D7_P D26
ADC0_out.CH.D7_N D27
GBTCLK0_M2C_P
GBTCLK0_M2C_N
LA01_P_CC
LA01_N_CC
LA05_P
LA05_N
LA09_P
LA09_N
LA13_P
LA13_N
LA17_P_CC
LA17_N_CC
LA23_P
LA23_N
LA26_P
LA26_N
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P1K
FMCSYNC_P
FMCSYNC_N
SPI.Addr0
SPI.MISO
ADC0_out.BDR.BDR_P G2
ADC0_out.BDR.BDR_N G3
ADC0_out.ADR.ADR_P G6
ADC0_out.ADR.ADR_N G7
ADC0_out.BH.D6_P G9
ADC0_out.BH.D6_N G10
FMCSYNC_P G12
FMCSYNC_N G13
ADC0_out.BH.D1_P G15
ADC0_out.BH.D1_N G16
ADC0_out.AH.D5_P G18
ADC0_out.AH.D5_N G19
ADC0_out.DH.D7_P G21
ADC0_out.DH.D7_N G22
ADC0_out.DH.D6_P G24
ADC0_out.DH.D6_N G25
G27
G28
ADC0_out.CH.D2_P G30
ADC0_out.CH.D2_N G31
ADC0_out.CH.D3_P G33
ADC0_out.CH.D3_N G34
ADC0_out.CH.D6_P G36
ADC0_out.CH.D6_N G37
CLK1_M2C_P
CLK1_M2C_N
LA00_P_CC
LA00_N_CC
LA03_P
LA03_N
LA08_P
LA08_N
LA12_P
LA12_N
LA16_P
LA16_N
LA20_P
LA20_N
LA22_N
LA22_P
LA25_P
LA25_N
LA29_P
LA29_N
LA31_P
LA31_N
LA33_P
LA33_N
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P1L
PRSNT_M2C_L
FMCADC_Rn
SPI.CSn
SPI.MOSI
SPI.SCK
SPI.Addr2
SPI.Addr1
PRSNT_M2C_L H2
ADC0_out.AH.D0_P H4
ADC0_out.AH.D0_N H5
ADC0_out.AH.D6_P H7
ADC0_out.AH.D6_N H8
ADC0_out.BH.D7_P H10
ADC0_out.BH.D7_N H11
ADC0_out.AH.D2_P H13
ADC0_out.AH.D2_N H14
ADC0_out.BH.D2_P H16
ADC0_out.BH.D2_N H17
ADC0_out.BH.D5_P H19
ADC0_out.BH.D5_N H20
FMCADC_Rn H22
H23
ADC0_out.DH.D4_P H25
ADC0_out.DH.D4_N H26
ADC0_out.CH.D5_P H28
ADC0_out.CH.D5_N H29
H31
H32
H34
H35
ADC0_out.CH.D4_P H37
ADC0_out.CH.D4_N H38
PRSNT_M2C_L
CLK0_M2C_P
CLK0_M2C_N
LA02_P
LA02_N
LA04_P
LA04_N
LA07_P
LA07_N
LA11_P
LA11_N
LA15_P
LA15_N
LA19_P
LA19_N
LA21_P
LA21_N
LA24_P
LA24_N
LA28_P
LA28_N
LA30_P
LA30_N
LA32_P
LA32_N
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SPI
SPI
MISO SPI.MISO
MOSI SPI.MOSI
SCK SPI.SCK
Addr0 SPI.Addr0
Addr1 SPI.Addr1
Addr2 SPI.Addr2
CSn SPI.CSn
I2C
I2C
eeprom_SCL I2C.EEPROM_SCL C30
eeprom_SDA I2C.EEPROM_SDA C31
eeprom_GA0 I2C.EEPROM_GA0 C34
eeprom_GA1 I2C.EEPROM_GA1 D35
SCL
SDA
GA0
GA1
P1C
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P1D
D29
D30
D31
D33
D34
TCK
TDI
TDO
TMS
TRST_L
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JTAG is not used on this board
ADCDigitalOutputs
ADC0_out
ADC0_out
ADR ADC0_out.ADR
AH ADC0_out.AH
BDR ADC0_out.BDR
BH ADC0_out.BH
CDR ADC0_out.CDR
CH ADC0_out.CH
DDR ADC0_out.DDR
DH ADC0_out.DH
FMC LPC & MGTs
File: FMCio_rowsCDGH.SchDoc
Size: Letter Revision: Rev 2
Date: 21-May-13 Sheet 19 of 21
Drawn by: Adam Gilbert
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ADCDigitalOutputs
ADR
AH
BDR
BH
CDR
CH
DDR
DH
ADC0_out
ADC1_out
ADC0_out.ADR
ADC0_out.AH
ADC0_out.BDR
ADC0_out.BH
ADC0_out.CDR
ADC0_out.CH
ADC0_out.DDR
ADC0_out.DH
ADC1_out.ADR
ADC1_out.AH
ADC1_out.BDR
ADC1_out.BH
ADC1_out.CDR
ADC1_out.CH
ADC1_out.DDR
ADC1_out.DH
GPIO
GPIO.ADCPLL_Lock
R120
680
1%
ADCPLL_LOCK_2v5
Level translate 3.3V to 2.5V
R121
2k
1%
FMC_GND
Have removed the over-range bit on ADC1
P1I
ADC1_out.DDR.DDR_P E2
ADC1_out.DDR.DDR_N E3
ADC1_out.CH.D5_P E6
ADC1_out.CH.D5_N E7
ADC1_out.DH.D0_P E9
ADC1_out.DH.D0_N E10
ADC1_out.DH.D4_P E12
ADC1_out.DH.D4_N E13
ADC1_out.DH.D7_P E15
ADC1_out.DH.D7_N E16
ADC0_out.CH.D1_P E18
ADC0_out.CH.D1_N E19
ADC1_out.BH.D2_P E21
ADC1_out.BH.D2_N E22
ADC1_out.BH.D4_P E24
ADC1_out.BH.D4_N E25
ADC1_out.AH.D0_P E27
ADC1_out.AH.D0_N E28
ADC1_out.AH.D4_P E30
ADC1_out.AH.D4_N E31
ADC1_out.BH.D5_P E33
ADC1_out.BH.D5_N E34
E36
E37
HA01_P_CC
HA01_N_CC
HA05_P
HA05_N
HA09_P
HA09_N
HA13_P
HA13_N
HA16_P
HA16_N
HA20_P
HA20_N
HB03_P
HB03_N
HB05_P
HB05_N
HB09_P
HB09_N
HB13_P
HB13_N
HB19_P
HB19_N
HB21_P
HB21_N
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P1J
ADC1_out.CDR.CDR_P F4
ADC1_out.CDR.CDR_N F5
ADC1_out.CH.D4_P F7
ADC1_out.CH.D4_N F8
ADC1_out.CH.D3_P F10
ADC1_out.CH.D3_N F11
ADC1_out.DH.D3_P F13
ADC1_out.DH.D3_N F14
ADC1_out.DH.D6_P F16
ADC1_out.DH.D6_N F17
ADC0_out.CH.D0_P F19
ADC0_out.CH.D0_N F20
ADC1_out.BH.D1_P F22
ADC1_out.BH.D1_N F23
ADC1_out.BH.D3_P F25
ADC1_out.BH.D3_N F26
ADC1_out.BH.D7_P F28
ADC1_out.BH.D7_N F29
ADC1_out.AH.D3_P F31
ADC1_out.AH.D3_N F32
ADC1_out.AH.D7_P F34
ADC1_out.AH.D7_N F35
F37
F38
HA00_P_CC
HA00_N_CC
HA04_P
HA04_N
HA08_P
HA08_N
HA12_P
HA12_N
HA15_P
HA15_N
HA19_P
HA19_N
HB02_P
HB02_N
HB04_P
HB04_N
HB08_P
HB08_N
HB12_P
HB12_N
HB16_P
HB16_N
HB20_P
HB20_N
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P1M
J2
J3
ADCPLL_LOCK_2v5 J6
FMCIO_Rn J7
ADC1_out.CH.D7_P J9
ADC1_out.CH.D7_N J10
ADC1_out.DH.D2_P J12
ADC1_out.DH.D2_N J13
ADC1_out.DH.D5_P J15
ADC1_out.DH.D5_N J16
ADC1_out.CH.D1_P J18
ADC1_out.CH.D1_N J19
ADC0_out.DH.D1_P J21
ADC0_out.DH.D1_N J22
ADC1_out.BH.D0_P J24
ADC1_out.BH.D0_N J25
ADC1_out.BH.D6_P J27
ADC1_out.BH.D6_N J28
ADC1_out.AH.D2_P J30
ADC1_out.AH.D2_N J31
ADC1_out.AH.D6_P J33
ADC1_out.AH.D6_N J34
Buck3v3Sync J36
Buck1v8Sync J37
CLK3_BIDIR_P
CLK3_BIDIR_N
HA03_P
HA03_N
HA07_P
HA07_N
HA11_P
HA11_N
HA14_P
HA14_N
HA18_P
HA18_N
HA22_P
HA22_N
HB01_P
HB01_N
HB07_P
HB07_N
HB11_P
HB11_N
HB15_P
HB15_N
HB18_P
HB18_N
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P1N
K4
K5
ADC1_out.CH.D2_P K7
ADC1_out.CH.D2_N K8
ADC1_out.CH.D6_P K10
ADC1_out.CH.D6_N K11
ADC1_out.DH.D1_P K13
ADC1_out.DH.D1_N K14
ADC1_out.CH.D0_P K16
ADC1_out.CH.D0_N K17
ADC0_out.DH.D0_P K19
ADC0_out.DH.D0_N K20
ADC0_out.DH.D2_P K22
ADC0_out.DH.D2_N K23
ADC1_out.ADR.ADR_P K25
ADC1_out.ADR.ADR_N K26
REFCLK_M2C_P K28
REFCLK_M2C_N K29
ADC1_out.AH.D1_P K31
ADC1_out.AH.D1_N K32
ADC1_out.AH.D5_P K34
ADC1_out.AH.D5_N K35
ADC1_out.BDR.BDR_P K37
ADC1_out.BDR.BDR_N K38
REFCLK_M2C_P
REFCLK_M2C_N
CLK2_BIDIR_P
CLK2_BIDIR_N
HA02_P
HA02_N
HA06_P
HA06_N
HA10_P
HA10_N
HA17_P_CC
HA17_N_CC
HA21_P
HA21_N
HA23_P
HA23_N
HB00_P_CC
HB00_N_CC
HB06_P_CC
HB06_N_CC
HB10_P
HB10_N
HB14_P
HB14_N
HB17_P_CC
HB17_N_CC
Samtec - ASP-134602-01
FMC HPC connector
File: FMCio_rowsEFJK.SchDoc
Size: Letter
Revision: Rev 2
Date: 21-May-13
Sheet 20 of 21
Drawn by: Adam Gilbert
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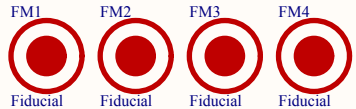
ADC 1 Analog input shield



ADC 0 Analog input shield



Fiducial marks



Logos

Logo1

LOGO

ChimeLogo1

Logo2

LOGO

ChimeLogo1

File: Mechanical.SchDoc	
Size: Letter Revision: Rev 2	
Date: 21-May-13 Sheet 21 of 21	
Drawn by: Adam Gilbert	

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File: ChimeLetterTemp.SchDot	
Size: Letter	Revision: Rev 2
Date: 21-May-13	Sheet of
Drawn by: Adam Gilbert	

McGill Cosmology

MGADC08

GHZ FMC ADC